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Rapid Communication

Hypothalamic paraventricular nucleus stimulation enhances c-Fos expression in spinal and supraspinal structures related to pain modulation

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ABSTRACT

The hypothalamic paraventricular nuclei (PVN) inhibits spinal nociception. Furthermore, projections from the PVN to other structures related to pain modulation exist, but a functional interaction has not yet been fully demonstrated. As an initial approach, we show here that PVN electric stimulation with the same parameters used to induce analgesia in rats enhances c-Fos expression not only in the dorsal horn of the spinal cord but also in the nucleus raphe magnus, locus coeruleus and the periaqueductal gray area. These results suggest that a functional interaction between these structures could occur, possibly to assure a mechanism of endogenous analgesia.

The role of the hypothalamic paraventricular nucleus (PVN) in endogenous analgesia has been suggested (DeLaTorre et al., 2009; Yang et al., 2009). Certainly, behavioral and electrophysiological approaches have shown that analgesia induced by PVN electric stimulation is mediated directly by oxytocin release from descending fibers from this nucleus to the dorsal horn of the spinal cord (Condés-Lara et al., 2006; Miranda-Cardenas et al., 2006; Rojas-Piloni et al., 2008). More recently, it was suggested that an interaction between the PVN and the locus coeruleus (LC) (Rojas-Piloni et al., 2012) or nucleus raphe magnus (RMg) (Condés-Lara et al., 2012) could have an additive antinociceptive effect.

The aim of the present study was to obtain more evidence about the functional interaction between the PVN and the LC and RMg structures related to pain modulation (Villanueva and Fields, 2004; West et al., 1993). Furthermore, we analyzed the possible interaction with the periaqueductal gray area (PAG), another structure involved in analgesia (Reynolds, 1969). For this purpose, we measured c-Fos expression (an indirect marker of neuronal activity) after PVN electric stimulation in the above structures. Our results show that PVN stimulation not only enhanced c-Fos expression in the dorsal horn of the spinal cord but also in structures related to inhibitory mechanisms of pain processing.

Experiments were carried out on male Wistar rats (280–310 g). The animals were housed individually in plastic cages under 12-h:12-h light/dark conditions in a special, temperature-controlled room ($22 \pm 2\,^{\circ}$C) with food and water *ad libitum*. All experimental procedures were approved by our Institutional Ethics Committee in accordance with the IASP ethical guidelines (Zimmermann, 1983). All efforts were made to limit distress and to use only the number of animals necessary to produce reliable scientific data.

Under anesthesia (ketamine/xylazine, 70/6 mg/kg, i.p.) the animals were placed in a stereotaxic apparatus. Then a small trephine was performed, and a concentric electrode (1 MΩ) was placed in the region corresponding to the PVN (accordingly to Paxinos and Watson, 1998). The animals were divided into 3 groups of 3 rats each. The first or stimulated (St) group received an electric stimulation consisting of 60 Hz pulses of 1-ms duration for 6 s, at 4 V and 300 μA; the second or sham-operated (SH) group, and a third group (control, Ctrl) without surgical manipulation. The parameters used to stimulate the PVN were taken from previous reports showing that this electric stimulation can inhibit the: (i) nociceptive behavior (Miranda-Cardenas et al., 2006); (ii) nociceptive

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responses of the dorsal horn wide dynamic range (WDR) neurons (Condés-Lara et al., 2008); and (iii) nociceptive transmission of spinothalamic tract and postsynaptic dorsal column neurons associated with nociceptive modulation (Rojas-Piloni et al., 2008).

Thirty minutes after PVN electric stimulation, the animals were deeply anesthetized and perfused through the heart using a peristaltic pump with a brief rinse of saline, followed by 4% paraformaldehyde in 0.1 M phosphate buffer (PB), pH 7.4. The brains and spinal cord were removed, stored in vials containing the same fixative for an additional 4 h, and cryoprotected in a 30% sucrose buffer solution for 2 days. The brains and spinal cords were cut in a freezing microtome at 40-μm intervals. Five adjacent series of sections were collected in 0.01 M phosphate-buffered saline (PBS). One series was processed later for Nissl (0.5% cresyl violet) and for immunohistochemical or immunofluorescent staining for c-Fos.

The c-Fos staining was carried out by incubating the sections in medium containing 0.01 M PBS with 0.5% Triton X-100 and blocking goat serum (10%) for 30 min at room temperature and then for 48 h at 4 °C with the primary antibody against c-Fos protein (Rabbit anti-fos 1:1000, Santa Cruz). After rinsing in PBS, the sections were: (i) incubated overnight with a biotinylated secondary antibody (Biotinylated goat anti-rabbit IgG 1:200, Vector) for immunohistochemistry; or (ii) incubated at 4 °C overnight with the fluorescent conjugated to the secondary antibody (anti-rabbit 1:300) for immunofluorescence. After that, the sections were processed with the avidin–biotin–peroxidase (ABC) kit (Vector Laboratories) for 4 h (for immunohistochemistry) at 37 °C. Bound antibody was then revealed with the nickel-diaminobenzidine (NiDAB) kit (Vector Laboratories). After rinsing, the sections were mounted on gelatin-coated glass slides, dehydrated, cleared, and coverslipped with Entellan™ (Merck KGaA, Darmstadt, Germany). The sections were observed with a Leica DMLB microscope and Leica DFC320 camera. A semi-automatic Image Processing System MD3 (Minnesota Datametrics) was used to count and plot the labeled cells. Only the rats with the electrode in the PVN were used (see Fig. 1B and C).

The data presented in the figures are the mean ± standard deviation of c-Fos-immunopositive cells that were counted for each structure, taking into account the sub-divisions and for the spinal cord, the lumbar segments. The differences between groups in the number of immunopositive cells were compared with a one-way analysis of variance, which was followed, if applicable, by the Student–Newman–Keuls *post hoc* test. Values of $p < 0.05$ were considered significantly different.

We processed 198 sections from the spinal cord between L2 and L5, resulting in a total of 43,478 cells with c-Fos. In the spinal cord c-Fos was mainly found in the stimulated group on the side ipsilateral to the PVN stimulation. In order to obtain more detailed information about the distribution of c-Fos, we divided the spinal dorsal horn into sections (L2–L5) in the lumbar region (Fig. 1D). Also, as shown in Fig. 1E and F, the sections from laminae I–II and III–V were analyzed separately. In all cases, we obtained a clear ipsilateral c-Fos activation when compared with the control or sham group.

In the case of the PAG we processed 188 sections, representing 4218 cells. Fig. 2 shows the effect on c-Fos expression after PVN electric stimulation. In contrast to the control (Fig. 2A) or sham (Fig. 2B), a significant increase in the number of c-Fos-immunopositive cells was observed in the stimulated group (Fig. 2C). For a more detailed analysis about the effect of PVN stimulation on this nucleus, the PAG was subdivided into its different regions (Fig. 2D). A clear activation was observed on both sides (ipsilateral and contralateral to PVN stimulation) in almost all regions.

In the LC we analyzed 66 sections, representing 117 c-Fos-positive cells. As shown in Fig. 3A, there was an increase in the number of immunoreactive cells in the stimulated group (Fig. 3A-3). Similar to the PAG result, we found bilateral activation (Fig. 3A-5). In the case of the RMg we examined 141 sections, in which we found 35 c-Fos-positive cells. Representative sections of the RMg are shown in Fig. 3-B. Although a significant difference ($p < 0.05$) was found between the ipsilateral and contralateral sides (Fig. 3B-5), expression of c-Fos was less than in the PAG and LC.

Taken together, the present research provides functional evidence about the possible relationship of the hypothalamic paraventricular nucleus (PVN) with certain structures related to pain modulation: the periaqueductal gray area (PAG), locus coeruleus (LC), and raphe magnus (RMg). Apart from the implications discussed below, and taken collectively, our findings show that, after PVN electric stimulation, the spinal cord was the main locus of c-Fos activation, possibly because the PVN projects directly to the dorsal horn. This result might also be due to the indirect activation of the PAG, LC, and RMg as well as other structures that were not explored in this study, but which also project to the spinal cord. These findings are relevant, because they support the notion that a functional antinociceptive interaction could exist between the PVN and structures related to pain modulation.

Regarding our experimental protocols, it is important to mention that apart from projections from the PVN to LC (Kobayashi et al., 1974), PAG and RMg (Larsen et al., 1996) the PVN extends widespread projections to structures (Geerling et al., 2010; Luiten et al., 1985; Magoun et al., 1938) not directly related to antinociception. However, as previously reported (Condés-Lara et al., 2008; Miranda-Cardenas et al., 2006; Rojas-Piloni et al., 2008), the parameters used in the PVN electric stimulation (60-Hz pulses of 1-ms duration for 6 s, at 4 V and 300 μA) to induce c-Fos expression are the same as those that induce analgesia in rats. Therefore, we consider that, under our experimental conditions, the expression of c-Fos could be associated with the antinociceptive effect induced by PVN stimulation.

Comparing the c-Fos activation between spinal cord and the supraspinal structures, the greater activation of this proto-oncogene is evident at spinal cord level. This difference could be associated with the fact that the parameters used to stimulate the PVN were chosen to observe selective inhibition of nociceptive transmission (Condés-Lara et al., 2008). In addition, the fact that we observed an ipsilateral activation in laminae I–II and III–V of lumbar segments L2–L5 (Fig. 1D–F) agrees with earlier studies (Kabat et al., 1935; Magoun et al., 1938), which assert the existence of a hypothalamic descending pathway reaching the spinal cord. Furthermore, direct projections from the PVN cells to lumbar and cervical segments of the dorsal horn of the spinal cord have been described (Condés-Lara et al., 2007). These results reinforce the idea that this PVN-spinal cord descending path is part of a diffuse and nonspecific system of pain modulation (Rojas-Piloni et al., 2008).

Considering that electric stimulation of the PAG produces a profound and selective analgesia (Reynolds, 1969), it is noteworthy that after PVN stimulation, a clear bilateral c-Fos expression was induced in the: (i) lateral (LPAG); (ii) dorsomedial (DMPAG); (iii) dorsolateral (DLPAG); and (iv) ventrolateral (VLPAG) portions of the PAG (Fig. 2). These results suggest that PVN activation is able to activate the PAG and therefore, apart from the direct oxytocinergic mechanism involved in the PVN-induced analgesia (Miranda-Cardenas et al., 2006), the role of the PAG and its descending projections to the spinal cord (Basbaum and Fields, 1978) could contribute in this analgesia.

In addition, a low immunoreactivity to c-Fos was found in the LC (Fig. 3A; bilateral) and RMg (Fig. 3B; ipsilateral). We could infer that this low immunoreactivity is due to the short time given to measure c-Fos after stimulation but it has been shown that 15 min

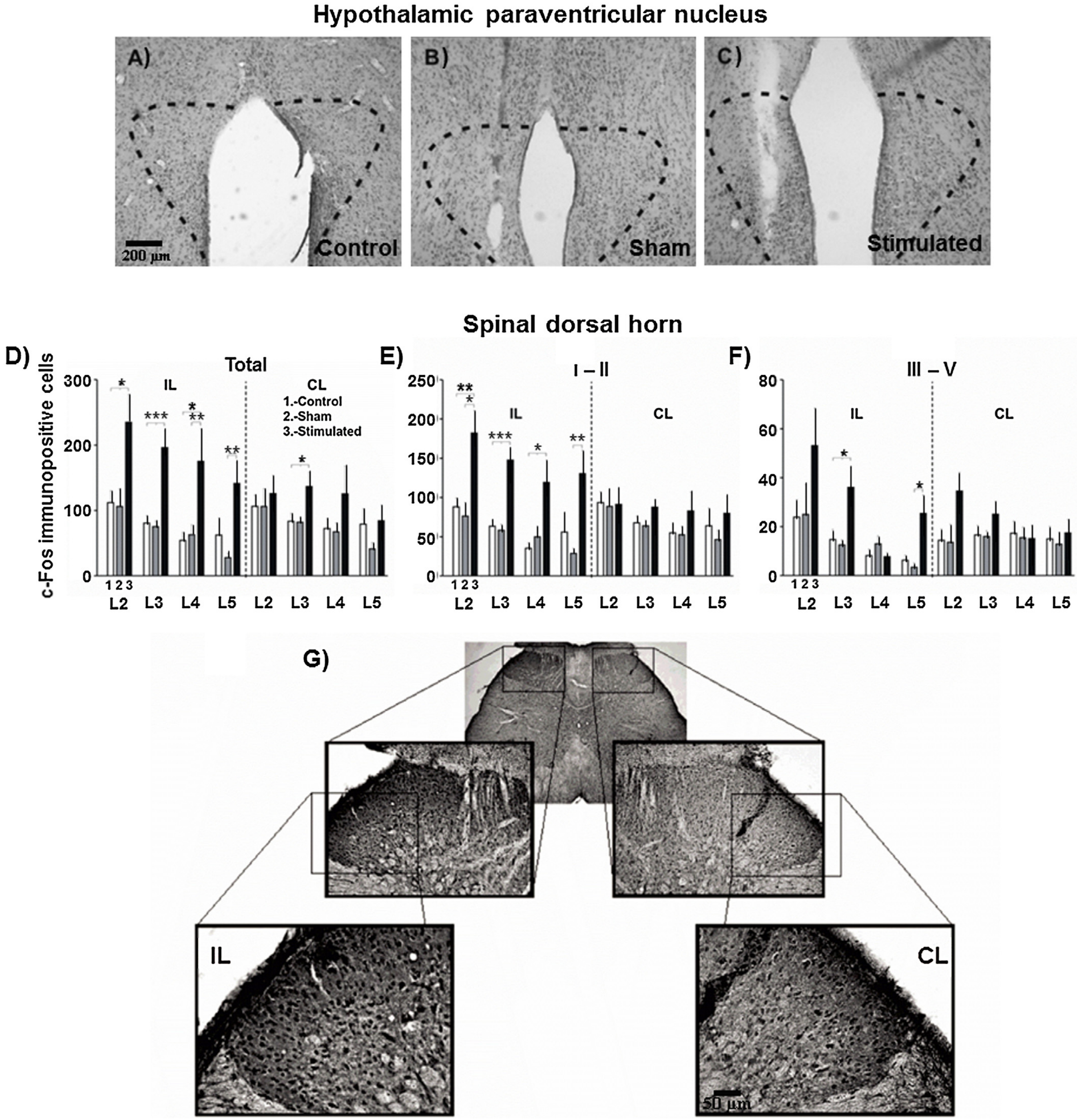

Fig. 1. Effect of the hypothalamic paraventricular (PVN) stimulation on c-Fos expression in the spinal dorsal horn. Panels (A)–(C) show a histological section of the PVN in control (A), sham (B) and stimulated (C) rats. (D)–(F) Show the number of c-Fos-positive cells on the ipsilateral (IL) and contralateral (CL) side of the spinal dorsal horn induced by paraventricular (PVN) electric stimulation. It is important to note that the expression of c-Fos was significantly enhanced on the ipsilateral side. Panel (D) shows photomicrographs with the histology of c-Fos-immunopositive cells. Abbreviations: IL, ipsilateral to PVN stimulation; CL, contralateral to PVN stimulation. The number of c-Fos-immunopositive cells shown in panels (D)–(F) represent the average number of positive cells per slide. The data is the mean ± standard deviation. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

after stimuli (*i.e.* chemical, electrical, mechanical) c-Fos immunoreactivity could be found in the central nervous system (Harris, 1998; Coggeshall, 2005; Dragunow & Robertson, 1987). In addition, it must be emphasized that c-Fos was not modified in the control and sham situations. Certainly, additional studies that fall beyond the scope of this investigation will be required to ascertain the time needed to induce maximal c-Fos immunoreactivity.

Admittedly, one limitation of our investigation could be the number of animals used ($n = 3$, each group), but we need to keep in mind that our study is based on the fact that projections from the PVN to LC, RMg and PAG exists (Kobayashi et al., 1974; Larsen et al., 1996). Furthermore, previous electrophysiological experiments showing an additive antinociceptive effect due to the interaction of the PVN with the LC (Rojas-Piloni et al., 2012) and

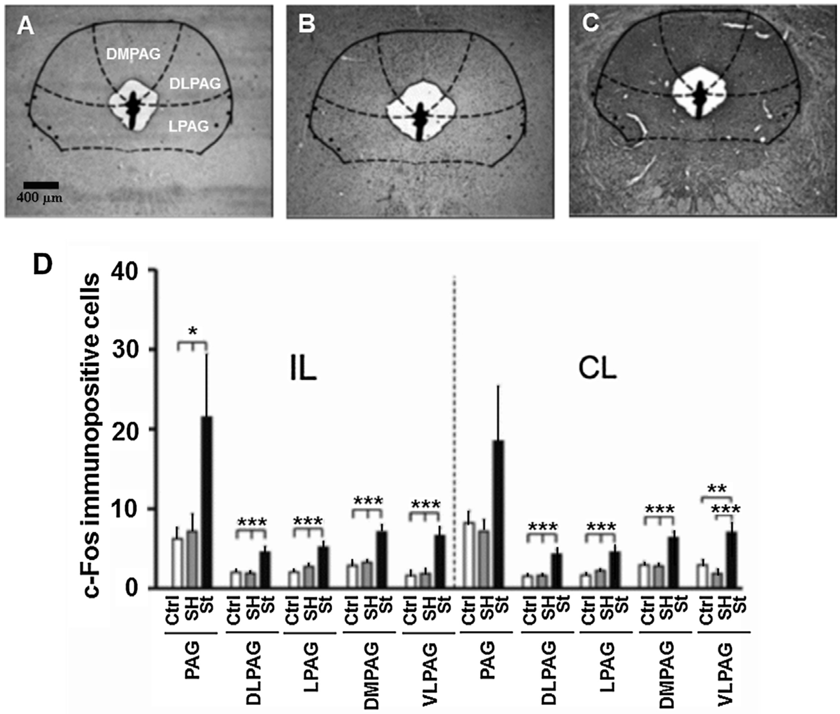

Fig. 2. Effects of the PVN stimulation on the number of c-Fos-positive cells in the periaqueductal gray area (PAG). The image shows photomicrographs of the: (A) control (Ctrl); (B) sham (SH); and (C) stimulated (St) rats. (D) A bilateral activation in most of the sub-nuclei was found. Abbreviations: DLPAG, dorsolateral; LPAG, lateral; DMPAG, dorsomedial; VLPAG, ventrolateral; Ctrl, control; Sh, sham; St, stimulated; IL, ipsilateral to PVN stimulation; CL, contralateral to PVN stimulation. The number of c-Fos-immunopositive cells in (D) is the average number of positive cells per slide. The data is the mean ± standard deviation. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

RMg (Condés-Lara et al., 2012). In any case, the objective of the present study is accomplished (*i.e.* this study shows that PVN electrical stimulation induces activation of these areas related to pain modulation).

Together, these results not only confirm the role of the PVN in modulating the neuronal activity of the spinal dorsal horn, but they also may suggest that it can activate other structures related to pain modulation. Certainly, descending pain inhibitory

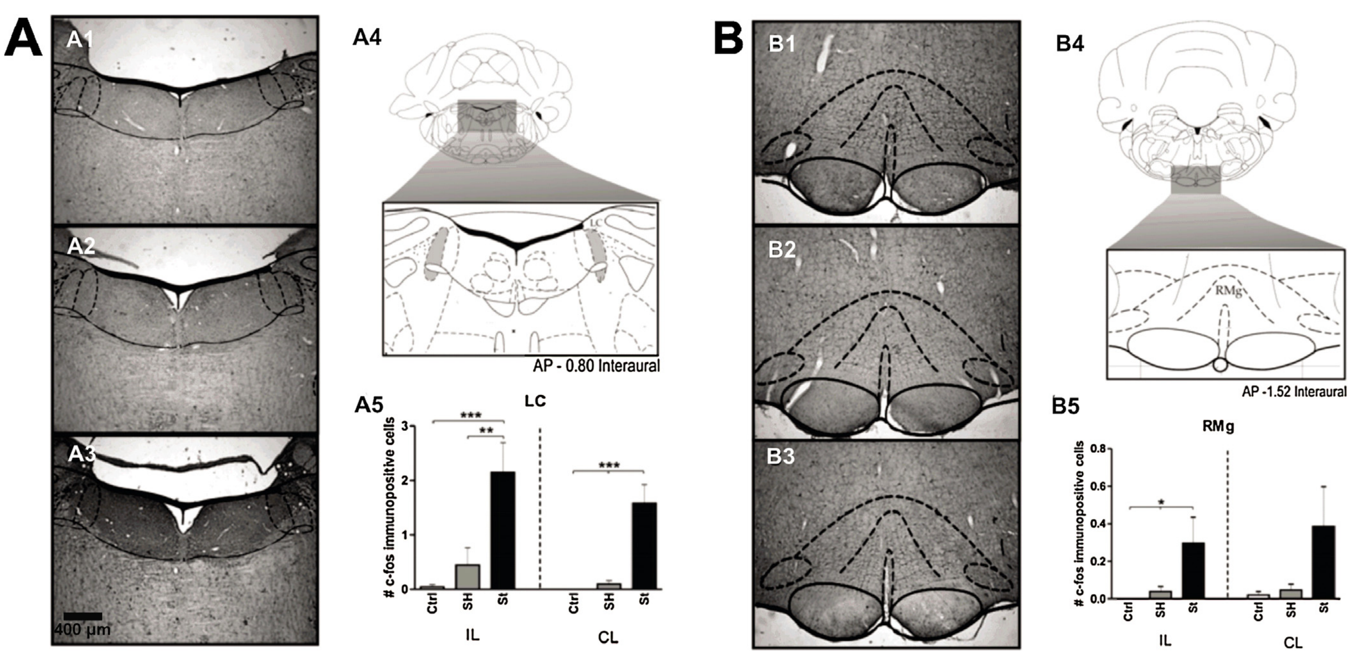

Fig. 3. Effects of the PVN stimulation on the number of c-Fos-positive cells in the (A) locus coeruleus (LC) and (B) raphe magnus (RMg). Histological sections from the control (A1 and B1), sham (A2 and B2), and stimulated (A3 and B3) groups are showed. Diagrams A4 and B4 taken from Paxinos and Watson (1998) show the area corresponding to the LC and RMg, respectively. Panel A5 shows a clear activation of the LC in the stimulated group, which led us to suggest a bilateral input from the PVN to the LC. Finally, panel B5 shows that, although minimal c-Fos expression was induced by PVN electrical stimulation, significant activation on the ipsilateral side was observed. IL, ipsilateral to PVN stimulation; CL, contralateral to PVN stimulation. The number of c-Fos-immunopositive cells shown in Panels A5 and B5 is the average number of positive cells per slide. The data is the mean ± standard deviation. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

mechanisms from these structures to dorsal horn of the spinal cord are relevant (Basbaum and Fields, 1978). The simplest interpretation of these findings suggests (but does not prove) that the activation of the LC, RMg and PAG by the PVN could be related to enhancement of the antinociception; however, the situation could be more complex than hypothesized here. Clearly, further experiments will be required to evaluate the neurophysiological impact of the c-Fos increase in these structures.

Lastly, it is interesting to note that oxytocin receptors (OTR) in these structures have been described (Hicks et al., 2012; Pagani et al., 2015; Yoshida et al., 2009). Furthermore, microinjection of oxytocin in the RMg (Wang et al., 2003) and PAG (Yang et al., 2011) decreases the pain threshold in rats. Admittedly, we have no direct evidence that OTR activation in our experiments is the responsible for the c-Fos increase. Nevertheless, it is important to keep in mind that the OTR signaling is mainly mediated by heterotrimeric $G_{q/11}$ proteins (Alexander et al., 2013) that, among other effects, enhance the adenylyl cyclase activity, a key step in the induction of the c-Fos protein (Nestler et al., 2009).

In conclusion, since PVN electrical stimulation increases c-Fos immunoreactivity at the PAG, LC and RMg, our results support the existence of an interaction between the PVN and these structures. A more detailed study about the functionality of the PVN as an orchestrator of endogenous analgesia in the *in vivo* situation is needed.

Conflict of interest

The authors declare that they have no commercial or financial relationship that could be construed as a potential conflict of interest.

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